

# The Effect of Musculoskeletal Pain on Motor Activity and Control

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**Abstract:** Aberrant movement patterns and postures are obvious to clinicians managing patients with musculoskeletal pain. However, some changes in motor function that occur in the presence of pain are less apparent. Clinical and basic science investigations have provided evidence of the effects of nociception on aspects of motor function. Both increases and decreases in muscle activity have been shown, along with alterations in neuronal control mechanisms, proprioception, and local muscle morphology. Various models have been proposed in an attempt to provide an explanation for some of these changes. These include the vicious cycle and pain adaptation models. Recent research has seen the emergence of a new model in which patterns of muscle activation and recruitment are altered in the presence of pain (neuromuscular activation model). These changes seem to particularly affect the ability of muscles to perform synergistic functions related to maintaining joint stability and control. These changes are believed to persist into the period of chronicity. This review shows current knowledge of the effect of musculoskeletal pain on the motor system and presents the various proposed models, in addition to other shown effects not covered by these models. The relevance of these models to both acute and chronic pain is considered. It is apparent that people experiencing musculoskeletal pain exhibit complex motor responses that may show some variation with the time course of the disorder.

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**Key words:** *Musculoskeletal pain, motor function, motor control, vicious cycle model, pain adaptation theory, neuromuscular activation model.*

Aberrant movement patterns and postures are obvious to and well identified by clinicians managing patients with musculoskeletal pain. Changes in motor activity and control observed in the clinic include increased activity in some muscle groups and inhibition or weakness of others. In addition, pain avoidance motor patterns, such as limping, decreased ranges of movement, listing of the spine, and loss of spinal curves, can be observed.<sup>1</sup> Some changes, however, are not so apparent. Motor control deficits in the form of muscle inhibition and altered patterns of muscle recruitment, resulting in the loss of joint control, recently have been recognized in such areas of the body as the lumbar

spine, cervical spine, and knee.<sup>2-4</sup> This loss of joint control may leave the subject biomechanically vulnerable to further injury or be the cause of ongoing pain.<sup>4</sup>

Experimental investigations in both animals and humans have provided evidence of the effect of nociception on various aspects of motor system function. Both increases and decreases in muscle activity have been shown under certain conditions. From these studies, various models have been proposed in an attempt to explain motor responses in the presence of pain. These include the vicious cycle<sup>5</sup> and pain adaptation models.<sup>6</sup> Although these models may go some way toward explaining the effect of pain on motor activity, the scenario that is presented to the clinician, particularly in the chronic pain situation, seems to be more complex and highlights the limitations of the current models. Recent research supports the emergence of a new model based on findings that patterns of muscle activation and recruitment are altered in the presence of pain.<sup>7-9</sup> Although possibly initiated in the acute stage of injury, these phenomena have been shown to persist into the period of chronicity.<sup>3,4,10,11</sup>

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Furthermore, the physiologic mechanisms responsible for pain-evoked motor responses are poorly understood, particularly at higher levels of the central nervous system. As Chudler and Dong<sup>12</sup> observe, little is known about the central mechanisms for integrating incoming nociceptive information that results in a motor response. It is apparent, however, that nociceptive input can have potent effects on the motor control functions of the central nervous system. Both peripheral and central phenomena have been proposed as explanations of the physiologic changes that occur in the motor system as a result of pain. However, none of these proposals fully explains the changes observed in the motor system in patients suffering musculoskeletal pain.

The purpose of this review is to show current concepts and knowledge of the effects of musculoskeletal pain on motor activity and to present evidence for an emerging model of altered patterns of neuromuscular control of muscles that perform key synergistic functions to stabilize the spine and major peripheral joints.

## Increased Muscle Activity

### *Flexor Withdrawal Reflex*

After tissue damage, there is a decrease in the threshold and an increase in the area of responsiveness and duration of response to mechanical stimuli in the normally highly stable flexor withdrawal reflex.<sup>13</sup> Strong C fiber input from skin and particularly from muscle has been shown to produce prolonged sensitization of spinal cord neurons in spinalized rats.<sup>14</sup> Intra-articular inflammation produced by the C fiber stimulant mustard oil also has been shown to produce a powerful, prolonged facilitation of the flexor withdrawal reflex.<sup>15,16</sup>

This injury-induced change to the flexor withdrawal reflex seems to depend on central nervous system changes rather than sensitization of peripheral afferents.<sup>17</sup> In investigations using intense electrical conditioning stimulation to a muscle nerve, exaggerated motor responses were evoked by stimuli applied to distant cutaneous nerves, including those to the opposite limb.<sup>16</sup> In addition, injury-induced changes were not abolished after a sensory block of the injury site with local anesthetic.<sup>17</sup> Because the response properties of both the central terminals of the primary afferents and the motor neurons themselves were normal, it was proposed that the hyperexcitability was due to changes in the interneurons that link the primary afferents with the flexor motor neurons.<sup>17</sup>

Although these effects are long lasting in decerebrate, spinalized preparations,<sup>18</sup> in spinally intact animals high-intensity, repetitive C fiber stimulation results in decreased reflex excitability.<sup>19</sup> Gjerstad et al<sup>20</sup> confirmed these findings in human experiments in which inhibition of cutaneomuscular reflexes occurred after experimental muscle pain. Andersen et al<sup>21</sup> showed both inhibition and facilitation of reflexes, depending on the site of induced muscle pain. These

reflex changes occurred only during the actual period of muscle pain and were not long lasting.<sup>21</sup> These findings are suggestive of the involvement of reflex modulation by descending inhibitory activity after intense C fiber input.<sup>21</sup> This is not to say that increased reflex excitability does not occur in human subjects but that ongoing spontaneous pain<sup>22</sup> and the involvement of convergent input (spatial summation)<sup>21</sup> may be required for its maintenance.

In addition to nociceptive input from muscle and joints, nociceptive input from meningeal tissue also produces aberrant motor responses. Hu et al<sup>23</sup> showed that application of a small fiber excitant and inflammatory irritant (mustard oil) to meningeal tissues resulted in increased electromyographic (EMG) activity in the neck and jaw muscles of rats. They noted especially large EMG responses in the trapezius muscle. They suggested that the most likely explanation for this increase in EMG activity was central sensitization of trigeminal second order nociceptive neurons.

Attempts have been made to extrapolate these findings to motor responses observed in specific clinical syndromes. Hall and Quintner<sup>24</sup> studied 4 subjects with upper limb radiculopathy. They showed increased EMG activity in the upper trapezius, deltoid, biceps, and triceps muscles after palpation of allodynic nerve tissue in each subject and widespread, increased EMG activity in response to elicitation of myotatic reflexes in the symptomatic upper limbs. They suggested that the widespread and multisegmental increase in EMG activity could represent a motor correlate of central sensitization. These changes could include increased excitability of alpha motor neurons and increased excitability of muscle spindles via gamma motor neurons.<sup>24</sup> As described previously, it is believed that facilitation of alpha motor neurons occurs indirectly via interneurons in the reflex pathway.<sup>17</sup> Because these interneurons have divergent connections,<sup>25</sup> their activation could explain the widespread nature of the EMG responses.<sup>24</sup> Abnormal responses of neurons within the dorsal horn (including prolonged afterdischarges to brief stimuli and ongoing spontaneous activity) could positively influence gamma motor neurons, leading to increased fusimotor drive and sensitization of spindles.<sup>24</sup>

Increased (possibly protective) muscle activity also has been shown in normal subjects when undergoing mechanical tests of neurodynamic function (such as the brachial plexus tension test described by Elvey<sup>26</sup>).<sup>27</sup> These studies support Sunderland's<sup>28</sup> proposal that muscle tone and activity protect the nervous system from tensile forces. It has been suggested that this increase in muscle activity is due to activation of the flexor withdrawal reflex.<sup>29,30</sup>

Although these effects of stimulating nerve tissue have been shown in normal subjects, Hall et al<sup>31</sup> showed that the flexor withdrawal response is more easily elicited in patients with chronic pain. They investigated hamstring muscle activity during the straight leg raise test for both normal subjects and subjects with

chronic L5 or S1 radiculopathy. Their results showed that hamstring muscle activity occurred before the onset of pain in radiculopathy subjects, whereas in normal subjects, a muscle response was only evident after pain was perceived.

These clinical studies have all used EMG as a measure of the muscle response to pain. Although EMG activity does not necessarily equate to the presence of pain,<sup>32</sup> increased EMG activity nevertheless does show a heightened response of the cutaneomotor reflexes in the presence of pain.

Changes in the behavior of spinal cord neurons (central sensitization) after a barrage of nociceptive input are believed to contribute to the sensory and motor responses observed clinically.<sup>33</sup> It has been proposed that if these central nervous system changes persist, a basis could exist for the continued presence of clinical signs (ie, increased EMG activity, aberrant movement and postures, and persistent pain) without any ongoing peripheral pathology.<sup>34</sup>

For central sensitization to be maintained, it is thought that some source of peripheral nociceptive input is required.<sup>35</sup> Suggestions for the source of this input have included the actions of protective or disused muscles or peripheral scar tissue.<sup>34</sup> In addition, it has been suggested that central sensitization can be maintained by subthreshold (initially nonpainful) mechanical stimuli.<sup>36</sup> This would be particularly likely in situations in which the phenotype of large diameter, myelinated afferent neurons had changed, allowing them to induce central sensitization in the same manner as unmyelinated C fibers.<sup>37</sup>

Therefore, input produced by sustained or repetitive non-noxious stimuli may maintain central sensitization and be a major source of posthealing pain and an ongoing stimulus for the changes in motor function seen in patients with chronic pain.<sup>34</sup>

### ***Modulation of Inhibitory Reflexes***

Recent studies have shown that pain may have an effect on inhibitory jaw reflexes. Painful heat stimuli and experimentally induced masseter muscle pain have been shown to result in modulation of the late suppression periods in human jaw-closing muscles.<sup>38-40</sup> These suppression periods are believed to be caused by inhibitory brain stem modulation elicited by electrical stimulation.<sup>39</sup> Although Wang et al<sup>39</sup> acknowledge that changes in the inhibitory reflexes were modest and as yet the clinical relevance of this is not clear, it nevertheless does provide further information of reflex changes in the presence of nociception.

### ***The Vicious Cycle Model***

In addition to increased EMG activity or muscle spasm being attributed to effects (albeit indirect) on alpha motor neurons, it has been suggested that excitability in gamma motor neurons may be a cause of increased muscle tension or spasm. Gamma or fusimotor neurons are influenced by the discharge of afferents from

joints,<sup>5</sup> muscle,<sup>41</sup> and skin.<sup>42</sup> If, via this system, afferent input has an important effect on motor control, changes in afferent input due to disease or injury may have important clinical consequences. However, the precise effect remains uncertain because of continuing debate as to the nature of the gamma influence, especially with respect to afferent input from muscle.

Some studies have suggested an increase in excitability of gamma motor neurons after muscle inflammation. Johansson and Sojka<sup>5</sup> observed that stimulation of group III and IV muscle afferents excites dynamic and static fusimotor neurons, thereby enhancing the sensitivity of primary and secondary muscle spindle afferents that determine muscle stiffness. They suggest that increased activity in primary muscle spindle afferents will then increase excitability in alpha motor neurons projecting to the primary muscle. They further hypothesize that this increased muscle stiffness may lead to increased metabolite production and further stiffness. In addition, increased activity in the secondary spindle afferents projects back onto the gamma system, perpetuating the enhanced muscle stiffness. They believe that these effects may be important in the generation and cause of occupational muscle tension (spasm) and pain<sup>5,43</sup> and have termed their model the *vicious cycle model*. It represents an expansion on a model originally proposed by Travell et al in 1942.<sup>44</sup> Further studies by Johansson's group have shown enhanced activity in primary and secondary spindle afferents after application of chemical mediators such as potassium chloride, lactic acid, bradykinin, and serotonin to muscle tissue.<sup>43,45,46</sup> In addition to altered responses after local injection into the ipsilateral muscle, these researchers also have shown modulation of secondary muscle spindle afferents after injection of bradykinin into the contralateral muscle.<sup>46</sup>

Excitatory effects on gamma motor neurons also have been shown in the neck muscles of the cat.<sup>47</sup> In this study, increased intramuscular concentration of bradykinin in the splenius and trapezius muscles induced large and long-lasting increases in the static stretch-sensitivity of muscle spindle afferents from ipsilateral and contralateral muscles, thereby lending support to the vicious cycle hypothesis of Johansson and Sojka.<sup>5</sup> In addition, they suggested that these findings could contribute to an understanding of the mechanisms behind reduced proprioception and balance and the high frequency of dizziness and vertigo in patients with chronic neck pain due to the interaction between altered proprioceptive input (from neck muscle spindles) and vestibular information.<sup>47</sup>

Input from articular afferents also has an effect on gamma motor neurons.<sup>48</sup> He et al,<sup>49</sup> in their experiments on the response of flexor motor neurons to knee joint inflammation in cats, found both inhibitory and excitatory effects on gamma motor neurons. Two thirds of the gamma motor neurons that developed increased reflex discharges, whereas one third of the neurons supplying the same muscle exhibited reduced reflex discharges. Generation of an inhibitory reflex response

may balance the influence of excitatory reflex responses in some motor neurons, thereby ensuring that the inflamed joint is kept in the midrange position where nociceptive joint afferents are less easily activated as a result of joint inflammation.<sup>50</sup> He et al pointed out that if the hypothesis of cocontraction is correct it would be expected that extensor muscles should also show a reduced reflex threshold. However, clinically, the picture is usually one of the development of flexion contractures with joint disease, suggesting a contraction of flexor muscles and relaxation of extensor muscles.<sup>49</sup> They suggested that changes in the reflex thresholds of both alpha and gamma motor neurons lead to excitation of flexor motor neurons and result in a flexor reflex pattern. Following on from this study, Johansson and Sjolander<sup>43</sup> proposed that afferent information from inflamed joints may act to further increase activity in the gamma muscle spindle system, thereby amplifying the vicious cycle model proposed earlier by these investigators.

However, the vicious cycle model has not been substantiated. Unfortunately, increased EMG activity that should be present with increased alpha motor neuron activity has not been shown.<sup>51-54</sup> Furthermore, an increase in the reflex threshold of gamma motor neurons in the gastrocnemius group and in the flexor muscle group (tibialis anterior) has been shown after artificially induced myositis in the gastrocnemius/soleus muscle in cats.<sup>41</sup> These investigators suggested a reduction in muscle tone as a possible functional implication of this finding to protect the damaged tissue. This could be an explanation for the apparent weakness of inflamed muscles. It has been speculated that fusimotor inhibition may be the reason for reflex atrophy of muscles in chronic musculoskeletal disorders.<sup>41</sup>

More recently, studies have investigated the influences of experimental muscle pain on spinal reflexes (namely the H reflex and stretch reflexes). The H reflex specifically relates to the excitability of alpha motor neurons, whereas the stretch reflex also involves muscle spindle dynamics.<sup>55</sup> The short latency stretch reflex has been shown to increase in both agonist and antagonist muscles.<sup>55</sup> However, because no H-reflex changes have been found after experimental pain, it has been suggested that a peripheral mechanism is involved, presumably the delta muscle spindle system.<sup>55,56</sup> However, without evidence of increased excitability of alpha motor neurons, the vicious cycle model cannot be supported.<sup>55</sup> Further investigations have shown that after experimentally induced muscle pain (in this case, soleus), facilitation of the stretch reflex occurs in the relaxed muscle only and not during functional activities such as sitting and walking.<sup>57</sup> However, the effects of pain on stretch reflexes remain unresolved because no increase in excitability after experimentally induced muscle pain in the lumbar spine has been reported.<sup>58</sup> Furthermore, although an increase has been measured with EMG, an inhibitory effect has been shown in single motor unit recordings.<sup>40</sup>

Although the validity of the vicious cycle model has been questioned, there is little doubt that tender, taut muscles can be found in patients with musculoskeletal pain. It has been proposed that this may be caused by endogenous muscle contracture or alteration in the viscoelastic properties of the muscle (nonelectrical) as opposed to contractile (electrical) changes.<sup>32</sup> The palpable taut band or trigger point makes the muscle feel tense but is not associated with propagated action potentials that would be identified as EMG activity.<sup>32</sup> Although this proposal may sound enticing to both patients and clinicians, the validity and reliability of the existence of trigger points have not been established.<sup>44</sup>

Therefore, it would seem that increased muscle activity may occur in the presence of pain via the flexor withdrawal reflex (central sensitization) but that ongoing nociceptive input may be required for it to be sustained. Because the presence of increased resting or postural EMG activity generally has not been shown, heightened flexor withdrawal responses may only occur with noxious provocation such as painful movements. It would seem that motor reflex changes in the presence of pain are dependent on whether the muscle is at rest or involved in static or dynamic activity.<sup>57</sup> Alternatively, painful, taut muscles palpated at rest in patients with musculoskeletal syndromes may be caused by changes in the viscoelastic properties of the muscles themselves and not to the contractile status of the muscles that could be measured by EMG.<sup>32</sup>

## Inhibition or Decrease in Muscle Activity

### *The Pain Adaptation Model*

Lund et al<sup>6</sup> refuted the vicious cycle model. They suggested that pain does not cause muscles to become hyperactive but that the ability to contract them is actually reduced. The exception to this is when a muscle is acting as an antagonist, in which case EMG activity has been shown to be higher than normal in the presence of pain. They argue that this is a useful reflex adaptation that limits range and velocity of movement, thereby reducing further injury and pain. This interaction between muscle pain and muscle coordination has been termed the *pain adaptation model*.<sup>6</sup> Support for the pain adaptation model has been shown by this group in studies involving experimentally induced pain in jaw muscles and the zygomatic periosteum in both animals<sup>59</sup> and humans.<sup>51</sup> In addition, other researchers have shown that maximal voluntary contraction is significantly reduced during saline-induced muscle pain.<sup>53</sup> This has also been shown clinically in patients with fibromyalgia.<sup>60</sup> It is suggested that these changes in motor output originate from alterations in the firing pattern of segmental interneurons in the spinal cord or brain stem.<sup>61,62</sup>

After saline injection into the gastrocnemius and tibialis anterior muscles, normal subjects showed altered gait coordination due to a decrease of EMG activity in the agonist muscle and an increase in activity in the antagonistic muscle.<sup>53</sup> In addition, saline-induced pain

in the posterior lumbar muscles and pathologic chronic low back pain result in increased activity in the lumbar paraspinal muscles during periods of the gait cycle in which, under normal circumstances, these muscles exhibit only minimal activity. Furthermore, reduced EMG activity is seen in subjects with experimental muscle pain during periods of the gait cycle in which these muscles normally exhibit maximal activity.<sup>63</sup> The investigators suggested that these findings correspond with the proposal of Lund et al<sup>6</sup> that the effect of pain on muscle function is dependent on the action of the muscle, and although it is difficult to determine when the back muscles act as agonists or antagonists, it seems that the presence of muscle pain affects the various phases of the gait cycle in different ways.<sup>53,63</sup> Further support for the pain adaptation model has been produced in studies in which experimental muscle pain was induced by saline injection into the human masseter muscle.<sup>64-66</sup> Observed changes included movements of smaller amplitude, slower movements, and reduced agonist EMG activity during pain. These motor effects may be explained by a facilitatory effect of activity in nociceptive muscle afferents on inhibitory brain-stem interneurons during agonist action.<sup>64-67</sup>

Therefore, it would seem that most published evidence questions the existence of muscle hyperactivity in the presence of pain, as postulated by the vicious cycle model. The pain adaptation model has been developed mainly from studies involving the induction of experimental muscle pain, which could be argued to be more akin to acute muscle pain in the clinical situation as opposed to chronic pain. Furthermore, this model primarily applies to pain from muscle and does not really account for pain arising from other tissue such as articular or nerve tissue or pain from an undetermined source (as occurs in many musculoskeletal syndromes). This model may have a more limited application in patients with chronic pain.

## Altered Patterns of Neuromuscular Activation

Recently, researchers have begun to investigate patterns of neuromuscular activation. It has been suggested that the presence of pain leads to inhibition or delayed activation of specific muscles or muscle groups that perform key synergistic functions. This produces alterations in the patterns of motor activity and recruitment during functional movement. It has been suggested that this inhibition usually occurs in the deep muscles local to the involved joint that perform a synergistic function to control joint stability.<sup>3,4,7,11,68</sup>

Evidence for altered patterns of neuromuscular control has been found in various studies. Nouwen et al<sup>69</sup> reported a relative increase in erector spinae activity and a decrease in lower abdominal activity during lumbar flexion in subjects with low back pain. Although the pain adaptation model may explain these findings, further studies have shown a more complex phenomenon. Selective fatigue of lumbar multifidus, as opposed

to other erector spinae muscles (longissimus and iliocostalis), has been detected by using EMG.<sup>70</sup> Therefore, although the net extensor torque remained the same, the synergistic influence of lumbar multifidus was diminished. Indeed some studies have identified potent changes in the lumbar multifidus in subjects with low back pain. Hides et al<sup>10</sup> used ultrasonography to show a marked atrophy of lumbar multifidus on the side ipsilateral to the patients' symptoms and also showed that these changes remained even after the patients had ceased to report pain.<sup>11</sup>

Hodges and Richardson<sup>3,7,8</sup> showed a delay in the onset of contraction of transversus abdominis in subjects with low back pain when they performed limb movements in various directions. This was in contrast to subjects without low back pain who showed that contraction of transversus abdominis precedes the contraction of the muscles producing the limb movement.<sup>3</sup> These investigators suggested that these findings indicate a deficit of motor control, resulting in inefficient muscular stabilization of the spine. Transversus abdominis is thought to provide rotational and lateral control to the spine.<sup>71</sup> This selective activation seems to be lost in individuals who have experienced low back pain. O'Sullivan et al<sup>4</sup> reported that subjects with low back pain (in this case spondylolysis or spondylolisthesis) have a different pattern of abdominal muscle activation when drawing in the abdominal wall muscles in supine crook lying compared with pain-free control subjects. In this study, subjects with chronic low back pain seem to substitute for dysfunction of the deep abdominal muscles by generating greater levels of activity in muscles, such as rectus abdominis, to dynamically stabilize the spine. They suggest that this may indicate a dysfunction of neuromuscular control because the synergistic function of the deep abdominal muscles is lost and substituted for by rectus abdominis.<sup>4</sup> Cholewicki et al<sup>68</sup> also have suggested that heightened activity in the superficial muscles may be a measurable compensation for loss of segmental spinal support.

In the cervical spine, it has been observed that the upper and deep cervical flexor muscles lose their endurance capacity in subjects with neck pain and headache.<sup>72,73</sup> The function of these muscles is considered vital for cervical segmental stability and postural control.<sup>74-76</sup> When testing for activity in these muscles, it has been noted that patients with neck pain tend to substitute with the superficial flexor muscles (sternocleidomastoid and scalenes) to achieve the desired position of the neck.<sup>9</sup> In addition to these muscles, it also is known that the posterior suboccipital muscles play an important role with respect to control of the head and neck.<sup>75,76</sup> Hallgren et al<sup>77</sup> and McPartland et al<sup>78</sup> noted atrophy in these muscles in patients with chronic neck pain. It is possible that the synergistic function of these muscles is lost and that other muscles, such as upper trapezius and levator scapulae, substitute for the suboccipital muscles during functional movements. Upper trapezius and levator scapulae have

been reported to show increased activity in subjects with neck pain.<sup>79</sup>

In both the lumbar and cervical spine, it seems that the dysfunctional muscles are the deep muscles with direct vertebral attachments. These muscles span the vertebrae and perform important synergistic supporting functions to stabilize articular segments rather than being primarily responsible for movement production.<sup>80</sup> It seems that although changes in the control of these muscles may be initiated in the presence of pain and tissue injury, they often are sustained beyond the acute pain phase and may contribute to the chronicity of many musculoskeletal problems. The neuromuscular activation model identifies dysfunction of synergistic muscle control as a specific and important consequence of pain and injury.

Although most investigations thus far have reported results from clinical studies, some evidence of altered patterns of motor recruitment has been shown after the induction of experimental muscle pain in upper trapezius.<sup>81</sup> In this study, shoulder muscle coordination was adversely influenced and reorganization of the pattern of muscle recruitment occurred after the induction of experimental muscle pain in the upper trapezius.<sup>81</sup>

### Neuronal Control Mechanisms

Apart from local neuromuscular control, other recent research has investigated more general neuronal control mechanisms in patients with musculoskeletal disorders. Byl and Sinnott<sup>82</sup> reported that subjects with low back pain were less likely to be able to balance on 1 foot with their eyes closed and showed significantly greater postural sway. Venna et al<sup>83</sup> observed that subjects with low back pain and neurologic leg deficits showed significantly slower reaction times in the upper limbs. Subjects with chronic low back pain also have been shown to have slower reaction times (for tasks such as using fingers to push a button in response to light stimuli) than pain-free control subjects, which these investigators suggest could be a result of pain.<sup>84</sup> Interestingly, inferior motor and patient handling skills have been recognized as risk factors for back injury in nurses and may be an important factor predisposing nurses to spinal pain.<sup>85</sup>

### Other Effects

#### Proprioceptive Deficits

Joint damage, which evokes abnormal afferent input and decreases gamma motor neuron excitability, also could decrease proprioceptive acuity.<sup>86</sup> Proprioceptive deficits have been shown in peripheral joints after traumatic injury,<sup>87</sup> chronic degenerative damage,<sup>88</sup> and experimentally induced muscle pain.<sup>89</sup> In addition to peripheral joints, changes also are evident in the cervical spine. Revel et al<sup>90</sup> found that cervicocephalic kinesthetic sensibility was significantly poorer in patients with chronic cervical pain than in normal subjects. Heikkila and Astrom<sup>91</sup> showed a loss of cervicocephalic kinesthetic awareness in patients after a whiplash

injury. They suggest that neck proprioception is affected by damage to the muscular and articular receptors or by alteration of afferent integration with respect to head orientation in space.<sup>91</sup> An increase in kinesthetic awareness may be important for pain control. Revel et al<sup>90</sup> showed that improved kinesthesia resulted in a reduction in neck pain in subjects with chronic cervical spine pain.

#### Pain/Reflex Inhibition of Muscle

Weakness and wasting of muscles acting around a damaged joint frequently are observed in clinical practice, and the phenomenon has been called *arthrogenous muscle weakness*.<sup>92</sup> After joint damage, immobilization or decreased activity due to pain may cause muscle fiber atrophy; however, this cannot explain all the muscle weakness that occurs immediately after injury.<sup>92</sup> There also is a reduction in voluntary muscle activation, which is defined as the failure to fully activate an uninjured muscle, which acts across an injured joint during a maximal voluntary contraction.<sup>92</sup>

This loss of maximal contraction may be caused by pain inhibition; however, it has been shown that inhibition still occurs when pain is not present.<sup>93</sup> Painless reflex inhibition is believed to occur when afferent stimuli from a joint reflexly inhibit the activation of alpha motor neurons in the ventral horn of the spinal cord.<sup>94</sup> Increased articular pressure due to joint effusion has been suggested as a mechanism for this inhibition. However, aspiration of a joint effusion still does not result in full muscle activation.<sup>93</sup> Hurley<sup>86</sup> suggested that reduced muscle activation may be caused by abnormal afferent input from damaged joint structures decreasing the excitability of alpha motor neurons. He showed that patients with coexisting anterior cruciate ligament ruptures and associated joint damage had a greater reduction in quadriceps activation than patients with isolated anterior cruciate ligament ruptures.

Most studies on arthrogenous muscle weakness have been conducted on knee joints in which quadriceps muscle weakness is apparent. However, reduced muscle activation is also likely to occur in other areas where joint damage has occurred. Hides et al<sup>10</sup> showed wasting of the lumbar multifidus muscle ipsilateral to the site of symptoms in patients with acute low back pain. McPartland et al<sup>78</sup> used magnetic resonance imaging to show that subjects with chronic neck pain had marked atrophy of the rectus capitis posterior major and minor muscles with significant fatty infiltration. Panjabi<sup>95</sup> suggested that any deterioration in muscle function, which may result from disuse, degeneration, disease, or injury, could cause inaccurate feedback to neuronal control systems, thereby affecting spinal joint control. It could therefore be possible that decreased muscle activation caused by either pain or reflex inhibition may have the potential to affect joint stability. This could have important implications for ongoing pain and dysfunction.

## Muscle Fiber Changes

Although this review deals mainly with neuronal control mechanisms, it is important to note that local muscle changes occur after tissue injury.<sup>96,97</sup> These changes also may contribute to impaired motor function and control in the presence of pain.

It has been suggested that joint injury or immobilization may lead to a change in the type or size of muscle fibers. Studies on muscle fiber size in the trapezius muscles of subjects with fibromyalgia have shown conflicting results, with some reporting atrophy of type II fibers, whereas others have found no change from control groups.<sup>98</sup>

Uhlig et al<sup>97</sup> investigated muscle fiber changes in ventral (sternocleidomastoid, omohyoid, and longus colli) and dorsal (rectus capitis posterior major, obliquus capitis inferior, splenius capitis, and trapezius) spinal muscles in patients who underwent surgery for cervical dysfunction of different etiologies (including rheumatoid arthritis, degenerative osteoarthritis, and post-traumatic instability). Signs of muscle fiber transformations were observed in all muscles investigated, as evidenced by an increased relative number of type IIC fibers. In the ventral muscles and obliquus capitis inferior, the occurrence of transformations correlated with the duration of symptoms. In the ventral muscles, the majority of transformations were in subjects with symptoms of short duration, whereas the obliquus capitis inferior showed fiber transformations in subjects with long symptom duration. In the other dorsal muscles, no correlation with duration of symptoms was found. Because a significantly higher percentage of type IIB fibers were found in muscles that had ceased transformation, these investigators suggest that the transformations occurred in the direction from "slow oxidative" to "fast glycolytic."<sup>97</sup>

Muscle fiber transformations occurred independently of the type of muscle (phasic or tonic), the gender and age of the patient, the type of condition, and the presence of neurologic deficit.<sup>97</sup> Because the changes occurred in all patients irrespective of their condition, the investigators suggested that neck pain itself must be considered the main stimulus for this response.<sup>97</sup>

These changes in slow twitch muscle fibers could be one reason for the clinical observation that the ventral deep neck flexor muscles show a poor static holding capacity or weakness.<sup>72</sup>

Histologic analysis of lumbar multifidus in subjects with lumbar disc herniation also has revealed structural changes. Mattila et al<sup>96</sup> reported the presence of core-targetoid or "moth-eaten" changes in type I fibers, which they suggested could be caused by denervation, muscle-spasm-induced ischemia, or excessive physical strain. Selective atrophy of type II muscle fibers also has been shown.<sup>96,99</sup>

## The Clinical Picture

Each of the phenomena outlined previously can be identified clinically during assessment of patients with

musculoskeletal pain. Some patients may exhibit increased activity in certain muscle groups with simultaneous inhibition or decreased activation in others. However, it would seem that none of the mechanisms discussed occurs in isolation and therefore does not provide a complete explanation of a patient's altered motor function in the presence of pain and subsequent to a painful episode.

Increased muscle activity certainly is observed clinically but rarely in isolation, and the vicious cycle model has not been supported by many studies that showed no increase in resting EMG activity.<sup>41,51,53</sup> Moreover, it has been suggested that there is no clear relation between increased gamma motor neuron drive and muscular hyperactivity albeit in decerebrate animals.<sup>100</sup>

The pain adaptation model would seem to apply more to acute muscle pain in which agonist activity is inhibited and antagonistic activity facilitated to protect damaged tissue and prevent further injury. However, the clinical picture, particularly in patients with chronic pain, is much more complex than this. Certainly in the acute pain situation, some movements are restricted in this pattern, but other phenomena also are occurring. In the chronic pain situation, patients may exhibit a relatively normal range of movement despite the presence of pain and evidence of altered motor control.

Recent research indicates that changes in activation patterns of muscles occur in the presence of pain, which result in muscles with a synergistic function being inhibited and other muscles substituting during functional movement.<sup>9,101</sup> Use of these superficial movement production muscles as substitutes for the deep local muscles could result in loss of active joint support and control. Ultimately, this may lead to ongoing pain or recurrent episodes of pain. There is also some suggestion that changes in motor control systems may occur before the onset of pain and predispose to the development of spinal pain.<sup>4</sup> The neuromuscular activation model seems to provide an important insight into the chronic state in which selective control and activation of specific muscles have been lost.

It is apparent that subjects suffering musculoskeletal pain have extremely complex motor responses that may show some variation with the time course of the disorder.

## Possible Neurophysiologic Mechanisms

It is clear that pain affects and disrupts neuromuscular control. However, the precise neurophysiologic basis for these changes remains unclear. Few investigators have hypothesized a mechanism for their findings. Factors such as reflex inhibition due to pain, effusion, or other afferent input may explain some loss of muscular control; however, there may also be supraspinal involvement resulting in loss of balance reactions, postural sway, reduced reaction times, and delay in the activation of postural control muscles associated with limb movements.

Brain structures processing afferent information are interlinked with those processing motor input in both

precentral and postcentral gyrus structures.<sup>102</sup> In addition, there is a close anatomic relation between somatosensory input and motor output in the primary cortices.<sup>103</sup> Andersson et al,<sup>104</sup> using positron emission tomography, showed that brain areas activated by capsaicin-induced peripheral pain include the anterior cingulate gyrus, the central sulcus, anterior insular cortex, middle frontal gyrus, and the supplementary motor area. Otte et al,<sup>105</sup> also using positron emission tomography, showed significant hypometabolism in the parieto-occipital cortex in subjects with a whiplash injury, which they suggested may be caused by activation of nociceptive afferents from the upper cervical spine. Weiller et al<sup>103</sup> investigated areas of brain activation during active movement of the upper limb. These areas included the supplementary motor cortex, the inferior parietal cortex, the basal ganglia, and the cingulate gyrus. It was apparent that active movement also activated some areas activated by nociceptive input. An ongoing barrage of nociceptive input, as in the chronic pain situation, may potentially affect motor output and control. The opposite may also occur, whereby active movement (exercise) could potentially modulate nociception.

Little is known about the central nervous system mechanisms responsible for integrating incoming nociceptive information that results in a motor response.<sup>12</sup> It is possible, however, that nociceptive input could disrupt central motor control mechanisms. Potentially, this could occur at various sites within the central nervous system.

## Summary and Conclusions

Musculoskeletal pain potentially produces many changes in motor activity. Some of these changes can be explained by peripheral mechanisms in the muscles themselves and by mechanisms within the central nervous system. Certainly, pain has a potent effect on motor activity and control.

The dysfunction that occurs in the neuromuscular system in the presence of pain is extremely complex. In addition to the more obvious changes, such as increased muscle activity in some muscle groups and

inhibition of others, more subtle anomalous patterns of neuromuscular activation seem to occur. Some of the models discussed (the vicious cycle and pain adaptation models), although occurring in patients with chronic pain, clinically seem to be more applicable in the acute pain situation. Loss of selective activation and inhibition of certain muscles that perform key synergistic functions, leading to altered patterns of neuromuscular activation (neuromuscular activation model) and the ensuing loss of joint stability and control, are initiated with acute pain and tissue injury. However, these phenomena persist into the period of chronicity and could be one reason for ongoing symptoms.

For optimal treatment, the assessment of patients with musculoskeletal pain therefore should include identification of altered neuromuscular activation as well as assessment of appropriate timing of muscle activation,<sup>7</sup> patterns of cocontraction,<sup>101</sup> and proprioceptive control.<sup>90</sup> Exercise programs aimed at training control of the deep muscles in the lumbar spine have shown some early success.<sup>4,11</sup> It is possible that this approach could be applied more widely in the re-education of other muscles that perform synergistic functions.

It is also possible that early intervention to reduce the barrage of nociceptive input may reduce potential motor control dysfunction. However, some changes in muscle function have been shown to occur within 24 hours of injury.<sup>10</sup> Therefore, early introduction of exercises aimed at maintaining motor control patterns may be of considerable importance.

It is imperative that further basic research is undertaken to investigate the neurophysiologic mechanisms responsible for these motor changes. At present, the central nervous system mechanisms that integrate nociceptive information into a motor response are not well defined and require further elucidation.

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